IGCSE 0607 Practice Questions — E1.6 — Combined Questions & Markscheme

The four operations

Total Questions: 18 — **Total Marks:** 99 Marks

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- **E1.6** The four operations
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 Work out the exact value of

$$-3 - 4 \times \left(\frac{1}{2} - \frac{3}{4}\right)^2$$

..... **[3]**

2 **(a)** The temperature at midnight was $-5°C$.
By noon, the temperature had risen by $14°C$.
Write down the temperature at noon.

..... **[1]**

(b) Work out the difference between $-8°C$ and $17°C$.

..... **[1]**

(c) Work out the value of

$$24 - 12 \div (3 - 7)$$

..... **[2]**

3 **(a)** Work out the value of

$$12 - 2 \times (-3) + 4^2$$

..... **[2]**

(b) Work out the value of

$$\frac{3}{4} + \frac{1}{6} \times \frac{3}{2}$$

Give your answer in its simplest form.

..... **[2]**

4 **(a)** Work out the value of

$$12 - 3 \times (-4) + (15 \div -3)$$

..... **[2]**

(b) Work out the value of

$$\frac{2}{5} \div \frac{4}{7} - \frac{1}{10}$$

Give your answer as a fraction in its simplest form.

..... **[2]**

5 **(a)** Work out the value of

$$15 - 3 \times (4 - (-2))$$

..... **[2]**

(b) Work out the value of

$$2\frac{1}{4} \div \frac{3}{8} - 1\frac{1}{2}$$

Give your answer in its simplest form.

..... **[3]**

6 **(a)** Work out the value of

$12 - 4 \times (5 - 9)$

..... **[2]**

(b) Work out the value of

$1.2 \div 0.04$

..... **[2]**

(c) The temperature at 04:00 was $-5.2°C$.
By 14:00, the temperature had risen by $8.7°C$.
Work out the temperature at 14:00.

..... °C **[1]**

7 **(a)** Work out the exact value of

$\frac{5}{6} + \frac{2}{3} \times \frac{9}{10}$ Give your answer as a mixed number in its simplest form.

..... **[3]**

(b) Work out the exact value of

$2\frac{1}{4} \div \left(1\frac{1}{8} - \frac{3}{4}\right)$

..... **[3]**

8 **(a)** Calculate the exact value of

$$2\frac{2}{5} \div \left(1\frac{1}{3} - \frac{5}{6}\right)$$

Show all your working and give your answer as a mixed number in its simplest form.

..... **[4]**

(b) Work out the exact value of

$$\frac{0.45 \times 1.2}{0.06}$$

..... **[3]**

9 Solve the equation, giving your answer as a fraction in its simplest form.

$$\frac{x+\frac{1}{3}}{2-0.4} = -1\frac{1}{4}$$

$x =$ ..... **[5]**

10 Work out the exact value of

$$\frac{2\frac{1}{3} \times 1\frac{2}{7} - \sqrt{\frac{25}{9}}}{\left(\frac{1}{2}\right)^3 \div \frac{3}{16}}$$

..... **[5]**

11 Without using a calculator, work out

$$\frac{3\frac{2}{3}+2\frac{1}{4}\times 1\frac{1}{3}}{4\frac{1}{2}-1\frac{5}{6}}$$

You must show all your working and give your answer as a fraction in its simplest form.

..... **[5]**

12 **(a)** Work out the value of

$$2\frac{2}{3} \div \left(1\frac{1}{4} - \frac{5}{6}\right)$$

Give your answer as a mixed number in its simplest form.

..... **[4]**

(b) A rectangular field has a length of $12\frac{1}{2}$ m and an area of $83\frac{1}{3}$ m^2.
Work out the perimeter of the field.
Give your answer as a mixed number in its simplest form.

..... m **[4]**

13 **(a)** At 06:00, the temperature in a city is $-4.5°C$.
By 14:00, the temperature has risen by $11.2°C$.
By 22:00, the temperature has dropped by $8.1°C$ from the temperature at 14:00.

Work out the temperature at 22:00.

..... **[2]**

(b) Use your calculator to work out the value of

$$\frac{4.72^2 - \sqrt[3]{18.5}}{3.1 \times (0.45 + 1.2)}$$

Give your answer correct to 3 significant figures.

..... **[2]**

14 A water tank has a capacity of 85.5 litres.
It is initially completely empty.
A hose pumps water into the tank at a rate of $\frac{3}{4}$ of a litre per minute.
At the same time, a leak in the tank loses water at a rate of 0.45 litres per minute.

Work out how long it takes to completely fill the tank.
Give your answer in hours and minutes.

hours ..... , minutes ..... **[4]**

15 The temperature of a liquid is $-4.5°C$.
It is heated so that its temperature rises by $2.8°C$ every minute for 15 minutes.

(a) Work out the temperature of the liquid after the first 15 minutes.

..... **[2]**

(b) After the 15 minutes, the liquid is cooled so its temperature falls by $1.2°C$ every minute for m minutes.
After the m minutes of cooling, the temperature of the liquid is $9.9°C$.
Work out the value of m.

$m =$ ..... **[3]**

16 A scientific freezer is used to change the temperature of liquid samples.

(a) Sample A has an initial temperature of 12.5°C.
It cools at a constant rate of 3.2°C per minute.
Calculate the temperature of Sample A after 8 minutes.

..... **[2]**

(b) Sample B has an initial temperature of −4.5°C.
To find its final temperature, the initial temperature is multiplied by −2, and the result is then decreased by 18.3°C.
Calculate the positive difference between the final temperature of Sample B and the final temperature of Sample A.

..... **[4]**

17 A large container has a total capacity of 1000 litres.

(a) The container is initially $\frac{2}{5}$ full of water.
Water is pumped into the container at a constant rate of 18.5 litres per minute for 20 minutes.
Work out the volume of water in the container after these 20 minutes.

..... litres **[3]**

(b) Water then leaks from the container at a constant rate of 4.5 litres per minute.
Calculate the time taken for the container to become exactly half full.

..... minutes **[3]**

(c) Once the container is half full, the leak is fixed.
A different pump then fills the container completely in 1 hour and 20 minutes.
Calculate the average rate at which this pump fills the container.
Give your answer in litres per minute.

..... litres per minute **[3]**

18 **(a)** A tank contains 450.5 litres of oil.
The oil is poured into bottles.
Each bottle holds exactly 1.25 litres.
Work out the number of completely filled bottles and the volume of oil left over.

Number of bottles = ... , Volume left over = ... **[3]**

(b) A machine pumps water at a rate of 1800 litres per minute.
It is used to fill a rectangular pool of length 24.5 m, width 12 m and depth 1.8 m.
($1\ m^3 = 1000$ litres)

Calculate the time taken to fill the pool completely.
Give your answer in hours and minutes.

hours, minutes **[4]**

(c) A freezer is set at $-18°C$.
During a power cut, the temperature inside the freezer rises by $1.5°C$ per hour.
The power cut lasts for 6 hours and 20 minutes.
Work out the temperature in the freezer when the power is restored.

..... °C **[3]**

1 $\left(-\frac{1}{4}\right)^2$ or $\frac{1}{16}$ **[M1]**

$4 \times \frac{1}{16}$ or $\frac{1}{4}$ **[M1]**

$-3\frac{1}{4}$ or $-\frac{13}{4}$ or -3.25 **[A1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** 9 **[B1]**

(b) 25 **[B1]**

(c) 27 **[B2]**

If 0 scored, M1 for -4 or -3 seen **[M1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** 34 **[B2]**

M1 for $-2 \times -3 = 6$ or $12 - (-6) + 16$ or $18 + 16$ seen

(b) 1 **[B2]**

M1 for $\frac{1}{6} \times \frac{3}{2} = \frac{3}{12}$ or $\frac{1}{4}$ seen

[Back to Question] *[Total Marks: 4]*

4 **(a)** Evaluates multiplication and division correctly to get $+12$ and -5 **[M1]**

19 **[A1]**

(b) Correct division seen (e.g. $\frac{2}{5} \times \frac{7}{4}$ or $\frac{14}{20}$ or $\frac{7}{10}$) **[M1]**

$\frac{3}{5}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $15 - 3 \times 6$ or $15 - 18$ **[M1]**

-3 **[A1]**

(b) $\frac{9}{4} \times \frac{8}{3}$ **[M1]**

6 **[A1]**

$4\frac{1}{2}$ or $\frac{9}{2}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 28 **[B2]**

If 0 scored, allow [M1] for correctly evaluating the bracket to -4 or for showing $12-(-16)$

(b) 30 **[B2]**

If 0 scored, allow [M1] for an equivalent fraction such as $\frac{120}{4}$ or $\frac{12}{0.4}$

(c) 3.5 **[B1]**

[Back to Question] *[Total Marks: 5]*

7 (a) $1\frac{13}{30}$ **[B3]**

If 0 scored, allow [M1] for correctly evaluating the multiplication as $\frac{18}{30}$ or $\frac{3}{5}$

Allow a further [M1] for finding a correct common denominator to add the fractions, e.g., $\frac{25}{30}+\frac{18}{30}$ or giving the improper fraction $\frac{43}{30}$

(b) 6 **[B3]**

If 0 scored, allow [M1] for correctly evaluating the bracket as $\frac{3}{8}$

Allow a further [M1] for converting to an improper fraction and correctly multiplying, e.g., $\frac{9}{4}\times\frac{8}{3}$

[Back to Question] *[Total Marks: 6]*

8 (a) $\frac{8}{6}-\frac{5}{6}$ or $\frac{1}{2}$ **[M1]**

$\frac{12}{5}\div$ their $\frac{1}{2}$ **[M1]**

$\frac{12}{5}\times\frac{2}{1}$ or equivalent multiplication **[M1]**

$4\frac{4}{5}$ **[A1]**

(b) 0.54 or $\frac{45\times 12}{600}$ or equivalent **[M1]**

$54\div 6$ or $0.54\div 0.06$ **[M1]**

9 **[A1]**

[Back to Question] *[Total Marks: 7]*

9 Denominator: $2-0.4=1.6$ or $\frac{8}{5}$ **[B1]**

Right hand side: $-1\frac{1}{4}=-\frac{5}{4}$ **[B1]**

$x+\frac{1}{3}=-\frac{5}{4}\times\frac{8}{5}$ **[M1]**

$x+\frac{1}{3}=-2$ **[A1]**

$x = -\frac{7}{3}$ or $-2\frac{1}{3}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

10 Numerator multiplication: $\frac{7}{3} \times \frac{9}{7} = 3$ **[M1]**

Evaluates root as $\frac{5}{3}$ and cube as $\frac{1}{8}$ **[M1]**

Denominator division: $\frac{1}{8} \times \frac{16}{3} = \frac{2}{3}$ **[M1]**

Numerator subtraction: $3 - \frac{5}{3} = \frac{4}{3}$ **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 Converts mixed numbers to improper fractions and multiplies correctly: $\frac{9}{4} \times \frac{4}{3} = 3$ (or $\frac{12}{4}$ or $\frac{36}{12}$) **[M1]**

Evaluates the numerator correctly: $3\frac{2}{3} + 3 = 6\frac{2}{3}$ (or $\frac{20}{3}$) **[M1]**

Evaluates the denominator correctly using a common denominator: $\frac{27}{6} - \frac{11}{6} = \frac{16}{6} = \frac{8}{3}$ **[M1]**

Sets up the correct division for the final step: $\frac{20}{3} \div \frac{8}{3} = \frac{20}{3} \times \frac{3}{8}$ (or $\frac{20}{8}$) **[M1]**

$\frac{5}{2}$ or $2\frac{1}{2}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

12 (a) $6\frac{2}{5}$ **[B4]**

M1 for changing mixed numbers to improper fractions, e.g., $\frac{5}{4}$ or $\frac{8}{3}$

M1 for writing fractions with a common denominator to subtract, e.g., $\frac{15}{12} - \frac{10}{12}$

M1 for $\frac{8}{3} \times \frac{12}{5}$ or $\frac{32}{5}$

(b) $38\frac{1}{3}$ **[B4]**

M1 for $83\frac{1}{3} \div 12\frac{1}{2}$

M1 for $\frac{250}{3} \times \frac{2}{25}$ or width $= \frac{20}{3}$ $(6\frac{2}{3})$

M1 for $2 \times \left(12\frac{1}{2} + \frac{20}{3}\right)$ or $2 \times \left(\frac{75}{6} + \frac{40}{6}\right)$

[Back to Question] *[Total Marks: 8]*

13 **(a)** Finds the temperature at 14:00 ($-4.5 + 11.2 = 6.7°C$) or sets up the full expression $(-4.5 + 11.2 - 8.1)$ **[M1]**

-1.4 **[A1]**

(b) Evaluates numerator to 19.63... or denominator to 5.115 **[M1]**

3.84 **[A1]**

[Back to Question] *[Total Marks: 4]*

14 Net rate of fill $= 0.75 - 0.45 = 0.3$ litres/min oe **[M1]**

Total time $= \frac{85.5}{0.3} = 285$ minutes **[M1]**

$285 \div 60 = 4$ with remainder 45 **[M1]**

4 hours 45 minutes **[A1]**

[Back to Question] *[Total Marks: 4]*

15 **(a)** 37.5 **[B2]**

M1 for 15×2.8 or 42 seen

(b) 23 **[B3]**

M1 for $37.5 - 1.2m = 9.9$ or $37.5 - 9.9$ or 27.6 seen

M1 for $27.6 \div 1.2$

[Back to Question] *[Total Marks: 5]*

16 **(a)** $12.5 - (3.2 \times 8)$ or $12.5 - 25.6$ **[M1]**

-13.1 **[A1]**

(b) $(-4.5 \times -2) - 18.3$ **[M1]**

Final temperature of Sample B $= -9.3$ **[A1]**

Difference $= |-9.3 - (-13.1)|$ or $13.1 - 9.3$ **[M1]**

3.8 **[A1]**

[Back to Question] *[Total Marks: 6]*

17 **(a)** $\frac{2}{5} \times 1000$ or 400 **[M1]**

18.5×20 or 370 **[M1]**

770 **[A1]**

(b) Target volume $= 500$ or volume to lose $= 770 - 500 = 270$ **[M1]**

$270 \div 4.5$ **[M1]**

60 **[A1]**

(c) Volume to fill $= 1000 - 500 = 500$ **[M1]**

1 hour 20 minutes = 80 minutes **[M1]**

$500 \div 80$ leading to 6.25 **[A1]**

[Back to Question] *[Total Marks: 9]*

18 **(a)** Number of bottles = 360 **[B1]**

Volume left over = 0.5 **[B2]**

If 0 scored for the volume left over, allow [M1] for $450.5 - (360 \times 1.25)$

(b) 4 hours 54 minutes **[B4]**

If 0 scored, allow [M1] for calculating the volume of the pool: $24.5 \times 12 \times 1.8 = 529.2$ m^3

Allow a further [M1] for converting the volume to litres: $529.2 \times 1000 = 529200$ litres

Allow a further [M1] for calculating total minutes: $\dfrac{529200}{1800} = 294$ minutes

(c) -8.5 **[B3]**

If 0 scored, allow [M1] for correctly interpreting 6 hours 20 minutes as a fraction or decimal of hours, e.g., $6\dfrac{1}{3}$ hours or $\dfrac{19}{3}$ hours

Allow a further [M1] for calculating the total temperature rise: $1.5 \times \dfrac{19}{3} = 9.5$

[Back to Question] *[Total Marks: 10]*